CUT Embedded Wall

CUT Embedded Wall is a specialized analytical–computational framework centered on advanced analytical earth-pressure formulations and their coupling with elastic beam-based soil–structure interaction analysis for embedded retaining walls.

Its philosophy is to combine:

- the transparency, robustness, computational efficiency, and reliability of analytical methods,
- with the realism and engineering confidence commonly associated with finite-element analysis.

In this way, CUT Embedded Wall seeks to provide FEM-level engineering insight without forcing the user into a heavy black-box numerical environment.

Highlights

- Analytical–computational platform for embedded wall analysis
- Powered by the Pantelidis Generalized Coefficients of Earth Pressure framework
- Unified treatment of at-rest, active, passive, and intermediate mobilized earth pressures
- Displacement-coupled mobilized earth-pressure formulations
- Cohesive–frictional soil modeling
- Static and pseudo-static seismic earth-pressure analysis

Wall Analysis Capabilities

- **Closed-form bending**
  Analytical elastic beam solution using closed-form bending formulations coupled with displacement-dependent mobilized earth-pressure distributions.

- **Differential equation**
  Advanced elastic beam model based on differential-equation soil–structure interaction analysis with displacement-coupled mobilized earth-pressure updating and nonlinear reinforcement/support interaction capabilities.

- **Base spring**
  Flexible embedded wall formulation combining differential-equation bending analysis with progressive rotational and translational base-spring release toward near free-base response. Includes bending, translation, rotation, nonlinear reinforcement/support interaction, and displacement-coupled mobilized earth-pressure updating.

- **Rigid wall**
  Rigid-body embedded wall model neglecting wall bending deformation while retaining displacement-dependent mobilized earth-pressure interaction through wall translation and rotation.

- **General case**
  Unified embedded wall framework combining bending deformation with rigid-body translation and rotation for complex staged wall–soil interaction problems involving displacement-coupled mobilized earth-pressure distributions.

Support and Reinforcement Systems

- Anchored walls
- Propped, braced, and strutted wall systems
- Multi-level support configurations
- True nodal reinforcement/support formulation
- Nonlinear one-way support interaction
- Modeling of support stiffness, capacity, spacing, free length, inclination, and prestress

Visualization and Engineering Workflow

- Real-scale geometry and reinforcement visualization
- High-quality engineering plots and animations

Core Concept

CUT Embedded Wall is built around the idea that many retaining wall problems can be treated with a high level of physical rigor **without abandoning analytical transparency**.

The software does not attempt to replace full finite element analysis in every possible situation. Instead, it provides a specialized analytical framework for earth pressure and embedded wall problems, where the governing mechanics are exposed, traceable, and computationally efficient.

This is particularly important because many conventional design-code approaches to earth pressures do not use a unified theoretical framework. They often rely on separate formulations for different states, intuitive correction factors, semi-empirical additions, and in some cases mathematical complications such as negative roots or nonphysical solution branches.

CUT Embedded Wall addresses these issues by using a unified earth pressure engine rather than a patchwork of disconnected expressions.

The Engine: Pantelidis Generalized Coefficients of Earth Pressure

At the heart of CUT Embedded Wall is the Pantelidis Generalized Coefficients of Earth Pressure framework.

This framework introduces a unified continuous formulation for describing the progressive mobilization of earth pressures, from at-rest conditions through intermediate mobilized states toward active or passive failure.

The framework allows continuous transition between:

- at-rest conditions,
- active mobilization,
- passive mobilization,
- and intermediate mobilized states.

It simultaneously incorporates:

- cohesion,
- friction,
- pseudo-static seismic loading,
- surcharge effects,
- and nonlinear mobilization behavior.

Why the K_XE Engine Is Important

The generalized coefficient engine is one of the central innovations of the program.

It offers a unified framework that:

- connects K_0, K_a, and K_p,
- includes cohesion,
- incorporates pseudo-static seismic loading,
- handles cohesive-frictional soils,
- avoids artificial switching between unrelated formulas,
- supports displacement-dependent mobilization,
- and provides a consistent basis for both retaining wall and pile problems.

This is a major departure from conventional earth pressure approaches, where active, passive, at-rest, cohesive, and seismic cases are frequently handled by separate equations or additional correction terms.

The engine, therefore, provides a coherent mechanical basis for a wide range of geotechnical problems involving earth pressures and soil reactions.

Validation of the Engine

The Pantelidis generalized earth pressure methodology has been extensively validated.

It has been checked against:

- hundreds of finite element models,
- independent numerical comparisons,
- centrifuge test data,
- seismic earth pressure benchmarks,
- retaining wall cases,
- axially loaded pile problems,
- laterally loaded pile problems.

This broad validation background is one of the main reasons the engine can be used as the central component of CUT Embedded Wall.

The same generalized coefficient concept has been extended beyond retaining walls to problems such as:

- embedded wall design,
- shaft resistance of axially loaded piles,
- laterally loaded piles,
- static and pseudo-static soil–structure interaction,
- and generalized lateral soil reaction problems.

Supported Analysis Configurations

CUT Embedded Wall currently supports the analysis of:

- cantilever embedded retaining walls,
- anchored retaining wall systems,
- propped and braced excavation systems,
- and multi-level reinforced wall configurations.

The framework is based on:

- flexible wall behavior,
- beam-type and differential-equation structural formulations,
- and displacement-coupled soil–structure interaction.

The wall response is calculated through:

  - bending-based structural analysis,

- iterative mobilization of earth pressures,

- and nonlinear reinforcement/support interaction.

The current implementation includes:

- fixed-base flexible wall analysis,
- nonlinear anchor interaction,
- nonlinear prop and strut interaction,
- support stiffness and capacity effects,
- prestress support modeling,
- support spacing, inclination, and geometric effects,
- and generalized reinforcement configurations within a unified computational framework.

The modular architecture also allows future extension toward additional analytical formulations, advanced soil–structure interaction models, and expanded reinforcement and boundary-condition capabilities.

Nonlinear Soil–Structure Interaction

CUT Embedded Wall couples:

- wall deformation,
- mobilized earth pressures,
- structural bending,
- and support forces.

The pressure field is not simply a fixed active or passive diagram. Instead, mobilized pressures are recalculated from the current displacement profile of the wall.

This allows the program to represent:

- partial mobilization,
- pressure redistribution,
- gradual support engagement,
- reduction of bending due to reinforcement,
- and the development of realistic deformation mechanisms.

Reinforcement and Support Modeling

Supports are not treated merely as pressure corrections.

Anchors, props, and struts are modeled as support elements interacting with the wall deformation.

The support model can include:

- axial stiffness,
- unsupported or free length,
- spacing,
- inclination,
- prestress,
- capacity,
- tension-only behavior,
- compression-only behavior,
- utilization checks,
- nodal reaction recovery.

This makes it possible to compare unsupported, anchored, and propped systems in a consistent way.

Engineering Visualization

The software produces engineering plots including:

- total earth pressure,
- net pressure,
- wall deflection,
- wall rotation,
- shear force,
- bending moment,
- K-diagram,
- normalized displacement / mobilization diagrams,
- convergence histories,
- support force tables.

Geometry and reinforcement are shown using true-scale visualization, so the wall depth, excavation level, and reinforcement lengths can be interpreted visually.

Streamlit Interface

The Streamlit version provides:

- clean and intuitive model input forms,
- editable engineering tables,
- interactive results dashboards,
- real-time engineering plots and visualization,
- automatic PDF report generation,
- CSV export capabilities,

- integrated manual/documentation access,
- modern browser-based engineering workflows,
- and cross-platform usability across Windows, macOS, Linux, iOS, and Android devices.

Intended Use

CUT Embedded Wall is intended for:

- teaching,
- academic research,
- method development,
- independent checking,
- parametric studies,
- comparison with finite element models,
- development of new geotechnical formulations.

It is especially useful where the engineer or researcher wants to understand the mechanics rather than simply obtain a black-box answer.

Related Papers

- Pantelidis, L. (2019). The generalized coefficients of earth pressure: a unified approach. Applied Sciences, 9(24), 5291.
- Pantelidis, L. From EN 1998-5: 2004 to prEN 1998-5: 2023: Has the calculation of earth pressures improved or deteriorated?
- Pantelidis, L. (2023, November). An Analytical Method for Designing Embedded Retaining Walls. In Rocscience International Conference (RIC 2023), pp. 217–224. Atlantis Press.
- Pantelidis, L. (2023, November). An Analytical Method for the Calculation of the Shaft Resistance of Axially Loaded Piles in Cohesive-Frictional Soils. In Rocscience International Conference (RIC 2023), pp. 209–216. Atlantis Press.
- Pantelidis, L. (2023, November). An Analytical Method for Designing Laterally Loaded Piles. In Rocscience International Conference (RIC 2023), pp. 567–574. Atlantis Press.
- Pantelidis, L. (2023). Designing laterally loaded piles relying on the Generalized Coefficient of Earth Pressure and static analysis of elastic beams. Research Square.
- Pantelidis, L., & Christodoulou, P. (2022). Comparing Eurocode 8-5 and AASHTO methods for earth pressure analysis against centrifuge tests, finite elements, and the Generalized Coefficients of Earth Pressure. Research Square, 3, 1–52.
- Pantelidis, L. (2022). Shaft resistance capacity of axially loaded piles in cohesive-frictional soils under static or pseudo-static conditions based on ground parameters.
- Pantelidis, L., Christodoulou, P., Assefa, E., & Sachpazis, C. (2021, November). The generalized coefficients of earth pressure: Numerical validation and comparison with

Eurocode 8-5 method. In International Conference on Mediterranean Geosciences Union, pp. 47–51. Springer Nature Switzerland.

Disclaimer

CUT Embedded Wall is educational and research software.

It is not a certified commercial design package. No warranty is provided regarding correctness, completeness, fitness for design, numerical stability under all conditions, or compliance with any specific design code.

All results must be reviewed and independently verified by qualified engineers before practical use.